

Piyush Sharma

New Delhi, Delhi

Education

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Thapar Institute of Engineering and Technology — BTech in Computer Science

Oct 2022 – Jun 2026

Experience

Software Engineer Intern | Starlight

Mar 2026 – Present

- Developed backend services for a full-stack PERN application, building REST APIs, relational database schemas, and asynchronous workflows for scalable media processing features.
- Built a Python-based image optimization pipeline converting images to WebP format with aspect-ratio preservation, reducing image sizes by 60–80% across multiple test datasets.

Software Engineer Intern | Synexoo

Jan 2025 – Mar 2025

- Built backend services for a social media platform using Node.js, Express.js, and MongoDB, implementing user profiles, follow systems, and optimized database queries.
- Developed a real-time messaging system using Socket.io with stable event handling and bidirectional communication support for concurrent active sessions.

Projects

MediaFlux | Flutter, FastAPI, Python

[GitHub](#) [APK](#)

- Built a full-stack AI-powered media optimization platform using Flutter and FastAPI supporting batch uploads, image compression, AVIF/WEBP conversion, resizing workflows, and real-time optimization tracking.
- Designed an asynchronous job-processing workflow with threaded background tasks, polling-based synchronization, and progressive result rendering for smoother user feedback during long-running operations.
- Implemented real-time frontend polling with animated result updates and auto-scrolling UI feedback across multiple optimization stages from upload to comparison generation.
- Developed media analysis pipelines generating heatmaps, comparison previews, compression metrics, and quality insights while preserving visual fidelity during optimization.
- Added backend workspace isolation, temporary file cleanup workers, and deployment workflows for reliable large-batch media processing.

SentinelAI | NestJS, Redis, Docker

[GitHub](#)

- Built an API security gateway using NestJS and Redis with adaptive risk scoring to classify incoming traffic and dynamically allow, throttle, or block suspicious requests.
- Implemented Redis-based sliding window rate limiting using sorted sets and TTL optimization, automatically detecting abnormal request bursts and reducing simulated abuse traffic during stress testing.
- Developed admin monitoring APIs for traffic inspection and manual IP controls; containerized backend services using Docker and deployed infrastructure workflows on Railway.

VCMtalker AI Assistant | Python, Ollama, Gemma, PyInstaller

[GitHub](#) [Download](#)

- Developed a fully offline desktop AI assistant using locally hosted LLMs (Ollama + Gemma) for natural-language interaction, OS automation, and file-based workflows with complete local data privacy.
- Built a voice interaction pipeline supporting application launching, web automation, and real-time system monitoring with reliable command execution across 50+ supported actions.

Skills

Programming Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Dart.

Computer Science Fundamentals: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Concurrency, Complexity Analysis.

Backend Engineering: Node.js, Express.js, FastAPI, NestJS, REST APIs, WebSockets, Redis, MongoDB, MySQL, SQLite, Rate Limiting, API Security.

Frontend & Mobile: Flutter, React.js, Next.js, Responsive UI Design, Real-time UI Rendering.

DevOps & Cloud: Docker, AWS, Git, Linux, Railway, Render, Replit, Vercel.

AI & Data: NLP, LLM Integration, Ollama, Image Optimization, WebP Compression, AVIF Conversion.

Competitive Programming

- **LeetCode** — Rating 1650+ (top 17.3%), solved 600+ problems across arrays, graphs, dynamic programming, greedy algorithms, and tree-based problems.
- **CodeChef** (3-star) — Rating 1600+, solved 200+ problems.